JGi JAIN DEEMED-TO-BE UNIVERSITY | ONLINE

Research Project

Semester-IV

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Date of Submission	29-July-2026

A study on the Impact of Business Intelligence Tools on Organizational Decision-Making in the Manufacturing Industry

Research Project submitted to Jain Online (Deemed-to-be University) In partial fulfillment of the requirements for the award of

Master of Business Administration

Submitted by

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Under the guidance of
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DECLARATION

I, *Deepanjali Haldar,* hereby declare that the Research Project Report titled Impact of Business Intelligence Tools on Organizational Decision-Making in the Manufacturing Industry has been prepared by me under the guidance of *Prof.Pradeep P.* I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of degree of Master of Business Intelligence & Analytics by Jain University, Bengaluru . I have undergone a project for a period of Eight Weeks. I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place:Bengaluru

Date:29-July-2026

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CERTIFICATE

This is to certify that the Research Project report submitted by Mr./Ms. *Deepanjali Haldar* bearing 241VMBR07775 on the title Impact of Business Intelligence Tools on Organizational Decision-Making in the Manufacturing Industry is a record of project work done by her during the academic year 2025-26 under my guidance and supervision partial fulfillment of Master of Business Intelligence & Analytics.

Place: Bengaluru

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ACKNOWLEDGEMENT

The Learners may acknowledge organization guide, University officials, faculty guide, other faculty members, and anyone else they wish to thank for their contribution towards accomplishing the research project successfully. The Learners may write in their own words and in small paragraph.

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EXECUTIVE SUMMARY

1. Overview & Sample Demographics:

This summary presents the key findings from a research survey conducted with 50 industry professionals evaluating the adoption, effectiveness, and implementation challenges of Business Intelligence (BI) tools across modern organizations.

- **Organizational Distribution**: The respondent base is strongly represented by **Large (46%)** and **Medium (44%)** enterprises, with Small enterprises making up 10%.
- I**ndustry Representation**: The Manufacturing sector represents the primary respondent group (**56%,** 28 respondents), followed by IT/Software (12%), Banking & Finance (10%), and Healthcare (10%).
- **Job Roles:** Responses were collected primarily from key decision-makers and analytics professionals, led by Managers / Team Leads (36%), Data Analysts (16%), Business Analysts (12%), and Operations Staff (10%).

2. BI Adoption & Tool Landscape:

BI technology has achieved near-universal penetration among surveyed organizations, driven by routine usage.

- **Adoption Rate**: **96%** of responding organizations actively utilize BI systems.
- **Market Share:** Microsoft Power BI dominates the organizational stack (64.6%, 31 users), followed by SAP BI (25%, 12 users). Tableau, Looker Studio, and Qlik Sense represent smaller niche shares.
- **Usage Engagement**: Engagement is high, with **51%** accessing BI platforms weekly and **40%** interacting with them daily.

3. Perceived Impact & Performed Metrics:

Respondents rated the impact of BI systems across several strategic dimensions on a 5-point Likert scale (where 1 = Strongly Disagree and 5 = Strongly Agree):

Dimension Evaluated	Mean Rating	Key Highlight
Overall Performance	4.90	Rated as the highest strategic outcome across
Business Trend	4.84	86% strongly agreed BI tools effectively
System Satisfaction	4.80	86% reported maximum satisfaction with
Report Readability &	4.79	High score indicating strong data
Productivity &	4.78	Consistent strong agreement on driving
Decision-Making	4.73	75.5% strongly agreed BI directly enhances

Net Promoter Score Indicator: **94%** of respondents explicitly recommended BI tools to other organizations, with the remaining 6% indicating “Maybe".

4. Implementation Challenges:

Despite high user satisfaction and business impact, organizational barriers remain during BI rollout:

1. **Lack of Skilled Personnel** (76% of responses): Cited as the single largest bottleneck during implementation.
2. **Resistance to Organizational Change** (34%): Internal adoption friction remains a secondary operational barrier.
3. **Security Concerns** (20%) & High Cost (18%): Governance, data privacy, and initial financial investments pose challenges for a subset of firms.
4. **Integration & Data Quality Issues** (22% combined): Technical hurdles relating to legacy software integration and data hygiene.

5. Conclusion & Recommendations

The research confirms that BI tools generate substantial value in operational efficiency, trend forecasting, and executive decision-making. To maximize ROI, organizations should prioritize employee upscaling programs and change management frameworks to resolve the talent gap and adoption friction that currently hinder seamless implementation.

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TABLE OF CONTENTS

Title	Page Nos.
Executive Summary	i
List of Tables	ii
List of Graphs	iii
Chapter 1: Introduction and Background	1-10
Chapter 2: Review of Literature	11-16
Chapter 3: Research Methodology	17-28
Chapter 4: Data Analysis and Interpretation	29-41
Chapter 5: Findings, Recommendations and Conclusion	42-45
References	
Annexures	

List of Tables

Table No.	Table Title	Page No.
1	Environmental Analysis (PESTEL Analysis)	8-9
2	Distribution of Respondents by Job Role	29
3	Distribution of Respondents by Organization Size	30
4	Does Your Organization Use BI Tools	31
5	Business Intelligence (BI) Tools Used in the Organization	32
6	Frequency of Using Business Intelligence (BI) Tools	33
7	Business Intelligence (BI) Improves Organizational Decision-Making	34
8	Business Intelligence (BI) Improves Operational Efficiency	35
9	Business Intelligence (BI) Increases Employee Productivity	36
10	Business Intelligence (BI) Reports are Easy to Understand:	37
11	Business Intelligence (BI) Helps Identify Business Trends and Patterns	38
12	Challenges Faced During Business Intelligence (BI) Implementation	39
13	Overall Satisfaction with Business Intelligence (BI) Tools	40
14	Recommendation to Adopt Business Intelligence (BI) Tools	41

CHAPTER 1

INTRODUCTION AND BACKGROUND

1. Purpose of the Study:

In the contemporary business environment, organizations across every sector are compelled to make decisions faster, with greater precision, and under conditions of increasing complexity and uncertainty. The manufacturing industry, in particular, generates enormous volumes of data every day through production lines, enterprise resource planning (ERP) systems, supply chain networks, quality control checkpoints, and customer interactions. However, the mere availability of data does not automatically translate into better decisions. Organizations require the ability to systematically collect, process, analyze, and visualize this data so that it can be converted into actionable insight. This is precisely the function performed by Business Intelligence (BI) tools, and it is this transformation from raw operational data to strategic insight that forms the central concern of the present study.

The purpose of this study is to critically examine how Business Intelligence tools, and specifically the combined application of SQL, Power BI, and Python, influence the quality, speed, and effectiveness of organizational decision-making within manufacturing enterprises. Manufacturing organizations occupy a unique position in this discussion because their decision-making spans multiple functional layers strategic decisions taken by top management regarding capacity expansion and market positioning, tactical decisions taken by middle management regarding production scheduling and supplier selection, and operational decisions taken on the shop floor regarding quality checks and machine utilization. Each of these layers depends on timely and accurate information, and BI tools are increasingly positioned as the technological backbone that supports this information flow.

More specifically, this study seeks to achieve the following purposes:

- To establish a clear conceptual understanding of what Business Intelligence means in a manufacturing context, and to distinguish it from related concepts such as data analytics, business analytics, and enterprise reporting.
- To examine, both theoretically and empirically, the mechanisms through which BI tools improve decision quality including improved data accuracy, faster access to information, better forecasting, and enhanced visualization of key performance indicators.
- To assess the tangible operational outcomes associated with BI adoption, such as gains in productivity, reductions in operational costs, improvements in inventory turnover, and better management of the supply chain.
- To document the practical challenges that manufacturing organizations encounter while implementing BI systems, including issues of data quality, system integration, cost of ownership, and resistance to change among employees.
- To demonstrate, through hands-on application of SQL, Power BI, and Python, how these tools function together across the analytics lifecycle from data extraction and cleaning to visualization and reporting thereby bridging the gap between theoretical understanding and technical competence.
- To generate evidence-based recommendations that manufacturing organizations, particularly small and medium enterprises operating in India, can use to plan, justify, and execute their own BI initiatives.

Ultimately, the purpose of the study extends beyond a purely academic exercise. It is intended to serve as a practical reference point for managers, analysts, and decision-makers who are evaluating whether, and how, to invest in Business Intelligence capabilities. By grounding the discussion in both established theory and current industry practice, the study aims to produce findings that are academically rigorous while remaining directly usable by practitioners in the manufacturing sector.

2. Introduction to the Topic:

Business Intelligence (BI) has become an essential component of modern organizations by enabling data-driven decision-making. With the rapid growth of digital technologies, organizations generate large volumes of data from daily business operations, customer interactions, production activities, and financial transactions. Business Intelligence transforms this raw data into meaningful information through data collection, integration, analysis, and visualization, helping organizations make informed and timely decisions.

The manufacturing industry is one of the largest contributors to economic growth and faces increasing pressure to improve productivity, reduce costs, and maintain product quality. Traditional decision-making methods based on manual reports and spreadsheets are often time-consuming and less efficient. The adoption of Business Intelligence tools has significantly improved the decision-making process by providing real-time insights into production, inventory, quality control, supply chain operations, and overall business performance.

Business Intelligence supports decision-making at strategic, tactical, and operational levels. Senior management uses BI for strategic planning and performance evaluation, while department managers rely on analytical reports to optimize resources and monitor key performance indicators (KPIs). Operational teams benefit from real-time dashboards that help identify production issues, reduce downtime, and improve process efficiency.

Modern BI solutions integrate advanced technologies such as cloud computing, artificial intelligence (AI), big data analytics, and the Internet of Things (IoT). These technologies enable organizations to analyze large datasets, predict future trends, monitor equipment performance, and automate business processes. As a result, manufacturing companies can make proactive decisions, improve operational efficiency, and enhance customer satisfaction.

Business Intelligence also plays a crucial role in quality management and supply chain optimization. By analyzing production and quality data, organizations can identify defects, reduce waste, improve product quality, and optimize inventory levels. Interactive dashboards and visual reports provide managers with accurate and timely information, enabling faster responses to operational challenges.

Despite its numerous advantages, the implementation of Business Intelligence involves challenges such as data integration, poor data quality, implementation costs, employee training, and resistance to organizational change. Addressing these challenges is essential to maximize the benefits of BI systems and ensure successful adoption across the organization.

This study focuses on the impact of Business Intelligence tools on organizational decision-making in the manufacturing industry. It also examines the practical application of SQL, Python, and Power BI in data extraction, analysis, visualization, and reporting. By evaluating the role of these technologies, the study aims to understand how Business Intelligence improves operational efficiency, supports better decision-making, and enhances overall organizational performance.

In conclusion, Business Intelligence has become a valuable strategic asset for manufacturing organizations. By transforming data into actionable insights, BI enables organizations to improve productivity, strengthen decision-making, and achieve sustainable competitive advantage in an increasingly competitive business environment.

3. Overview of Theoretical Concepts:

Business Intelligence (BI) is a technology-driven process that enables organizations to collect, integrate, analyze, and present business data to support effective decision-making. It combines various tools, technologies, and methodologies to transform raw data into meaningful information, allowing managers to make informed strategic, tactical, and operational decisions. The primary objective of Business Intelligence is to improve organizational performance by providing accurate, timely, and actionable insights.

One of the key concepts of Business Intelligence is the **Decision Support System (DSS)**. A Decision Support System assists managers in solving complex business problems by analyzing data and generating reports, forecasts, and recommendations. Unlike traditional reporting systems, DSS enables organizations to evaluate multiple alternatives before making decisions, thereby improving the quality and speed of decision-making.

Another important concept is **Data Warehousing**, which refers to the centralized storage of data collected from multiple sources such as Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) software, production databases, and financial systems. A data warehouse stores historical and current business data in a structured format, enabling efficient reporting and analysis. It provides a single source of truth for organizations, ensuring consistency and reliability in decision-making.

Data Analytics is another essential component of Business Intelligence. It involves examining, cleaning, transforming, and interpreting data to identify patterns, trends, and relationships. Data analytics can be classified into descriptive, diagnostic, predictive, and prescriptive analytics. While descriptive analytics explains what has happened, predictive analytics forecasts future outcomes, and prescriptive analytics recommends appropriate actions based on data insights.

Data Visualization plays a significant role in Business Intelligence by presenting complex information through charts, graphs, dashboards, and KPI indicators. Visualization tools such as Microsoft Power BI help users understand business performance quickly, identify trends, and communicate insights effectively. Interactive dashboards allow managers to monitor key performance indicators in real time and make faster business decisions.

The study also considers the role of **Key Performance Indicators (KPIs)** in measuring organizational performance. KPIs are measurable values used to evaluate how effectively an organization achieves its business objectives. In the manufacturing industry, common KPIs include production yield, Overall Equipment Effectiveness (OEE), inventory turnover, rejection rate, machine downtime, order fulfillment rate, and production efficiency. Monitoring these indicators enables organizations to identify performance gaps and implement corrective actions.

Another relevant theoretical concept is **Data-Driven Decision-Making (DDDM)**. This approach emphasizes making business decisions based on factual evidence and analytical insights rather than intuition or assumptions. Organizations that adopt data-driven decision-making improve operational efficiency, reduce business risks, and enhance overall organizational performance.

Finally, this study focuses on the practical application of **SQL, Python, and Power BI** within the Business Intelligence framework. SQL is used for extracting and managing data from databases, Python supports data cleaning, analysis, and automation, while Power BI enables interactive reporting and visualization. Together, these tools provide an integrated platform for analyzing business data and supporting organizational decision-making.

In summary, the theoretical concepts discussed in this study provide the foundation for understanding how Business Intelligence supports manufacturing organizations. By integrating data management, analytics, visualization, and decision-support systems, BI enables organizations to improve operational efficiency, optimize resources, and achieve sustainable competitive advantage through informed and data-driven decision-making.

4. Company/ Domain / Vertical /Industry Overview:

This research is conducted in the context of the **manufacturing industry**, with a specific focus on the application of **Business Intelligence (BI) tools** in improving organizational decision-making. The manufacturing sector plays a vital role in economic growth by producing goods, creating employment opportunities, and contributing significantly to national GDP. With increasing competition and rapid technological advancements, manufacturing organizations are adopting digital technologies to enhance operational efficiency, improve product quality, and achieve sustainable growth.

The manufacturing industry involves various activities such as procurement of raw materials, production planning, inventory management, quality control, logistics, and distribution. These processes generate a large amount of operational data on a daily basis. Effectively managing and analyzing this data is essential for improving productivity, reducing costs, and making informed business decisions. Business Intelligence tools help organizations convert this data into meaningful insights through data analysis, visualization, and reporting.

The domain of this study is **Business Intelligence and Analytics**, which focuses on collecting, processing, and analyzing business data to support decision-making. BI combines technologies such as data warehousing, data analytics, dashboards, and reporting tools to provide accurate and real-time

information. Organizations use these insights to monitor key performance indicators (KPIs), identify trends, forecast future outcomes, and optimize business operations.

The vertical considered in this research is the **manufacturing sector**, which includes industries such as automotive, electronics, machinery, pharmaceuticals, consumer goods, and industrial equipment. Manufacturing companies increasingly rely on data-driven strategies to improve production efficiency, minimize downtime, optimize inventory levels, and strengthen supply chain management. The adoption of Business Intelligence enables these organizations to respond quickly to market changes and customer demands while maintaining high standards of quality and operational excellence.

In recent years, digital transformation initiatives such as Artificial Intelligence (AI), the Internet of Things (IoT), cloud computing, and big data analytics have accelerated the adoption of Business Intelligence solutions in manufacturing. These technologies enable organizations to collect real-time production data, monitor equipment performance, predict maintenance requirements, and automate reporting processes. As a result, managers can make faster and more informed decisions based on accurate and up-to-date information.

In this study, the practical application of **SQL, Python, and Microsoft Power BI** is also emphasized. SQL is used for extracting and managing data from databases, Python supports data cleaning and analysis, and Power BI provides interactive dashboards and visual reports. Together, these tools help organizations transform raw data into actionable insights that improve business performance and support strategic decision-making.

Overall, the manufacturing industry provides an ideal environment for studying the impact of Business Intelligence on organizational decision-making. As organizations continue to embrace digital technologies, Business Intelligence has become an essential tool for improving efficiency, enhancing productivity, and maintaining a competitive advantage in today's dynamic business environment.

5. Environmental Analysis (PESTEL Analysis):

To develop a comprehensive understanding of the external environment within which Business Intelligence adoption occurs in the Indian manufacturing sector, this section applies the PESTEL framework, examining Political, Economic, Social, Technological, Environmental, and Legal factors that influence both the manufacturing industry broadly and the specific adoption of BI tools within it.

Table no:1

Factor	Analysis
Political	Government initiatives such as "Make in India," Production Linked Incentive (PLI) schemes, and the broader Digital India program actively encourage the modernization and digitalization of manufacturing operations. Trade policy, import export regulations, and shifting geopolitical dynamics including global supply chain realignment away from single-country dependence are pushing manufacturers to invest in better data visibility and analytics to manage more complex, distributed supply chains. Political stability and consistent policy support for digital infrastructure remain important enablers of continued BI investment.
Economic	India's manufacturing sector operates within a cost-sensitive environment where margins are often narrow, making the return-on-investment case for BI tools an important consideration, particularly for small and medium enterprises. Rising input costs, currency fluctuations, and competitive pressure from other low-cost manufacturing economies increase the incentive to use BI for cost optimization, waste reduction, and efficient resource utilization. At the same time, the falling cost of cloud computing and the availability of subscription-based BI platforms (rather than large upfront capital expenditure) is making analytics more economically accessible to a wider range of manufacturers.

Social	There is a generational and cultural shift underway within manufacturing organizations, as younger, more digitally literate managers and engineers enter the workforce and are generally more comfortable engaging with dashboards, data visualization, and analytics tools than earlier generations of shop-floor management. However, this is counterbalanced by a persistent skills gap — many existing employees lack formal training in data analysis or BI software, which contributes to the resistance and slow user adoption frequently cited as an implementation challenge. Organizational culture, particularly the extent to which leadership genuinely values data-driven decision-making over intuition-based decision-making, is also a significant social factor shaping BI adoption outcomes.
Technological	This is the most directly relevant PESTEL dimension for the present study. The proliferation of affordable cloud infrastructure, the maturity of tools such as SQL-based database systems, Python, and Power BI, and the growing integration of IoT sensors on the manufacturing shop floor have collectively lowered the technical barriers to BI adoption. At the same time, technological factors also present challenges including the complexity of integrating BI tools with legacy ERP and production systems, concerns over data quality and standardization across disparate systems, and the pace at which new analytical capabilities (such as AI-assisted analytics) continue to emerge, requiring ongoing organizational learning and adaptation.
Environmental	Sustainability considerations are increasingly shaping manufacturing operations, with growing regulatory and market pressure to reduce energy consumption, minimise waste, and lower carbon emissions. BI tools play a supporting role here by enabling organizations to monitor resource utilisation, track energy consumption patterns, and identify inefficiencies in production processes that contribute to environmental impact. As environmental, social, and governance (ESG) reporting requirements expand, the ability to accurately collect and report environmental performance data through BI systems is becoming an additional driver of adoption.
Legal	Manufacturing organizations operating in India must comply with a range of legal and regulatory requirements, including labour laws, industry-specific quality and safety standards, and data protection regulations such as the Digital Personal Data Protection Act. As BI systems increasingly consolidate sensitive operational, financial, and employee-related data into centralized dashboards, organizations face growing legal responsibility to ensure data security, privacy compliance, and proper governance of how business data is stored, accessed, and used factors that must be considered when planning and implementing BI initiatives.

Taken together, the PESTEL analysis indicates that the external environment for Business Intelligence adoption in Indian manufacturing is, on balance, favorable and increasingly supportive driven by government policy, falling technology costs, and competitive pressure but is simultaneously constrained by economic caution among smaller firms, social and skills-related resistance, and the technical complexity of integrating BI tools with existing legacy systems. These environmental factors provide important context for interpreting the primary research findings presented in later chapters of this study, and help explain why BI adoption levels vary so considerably across the Indian manufacturing landscape.

CHAPTER 2

REVIEW OF LITERATURE

Domain/ Topic Specific Review:

Introduction

Business Intelligence (BI) has become one of the most important technologies for organizations seeking to improve decision-making and operational performance. In today's competitive business environment, organizations generate large volumes of data from production, sales, inventory, finance, and customer interactions. Converting this data into meaningful information is essential for making timely and effective business decisions. Business Intelligence enables organizations to collect, integrate, analyze, and visualize data, allowing managers to monitor business performance and identify opportunities for improvement.

The manufacturing industry has experienced significant digital transformation in recent years through the adoption of technologies such as Business Intelligence, Big Data, Artificial Intelligence (AI), and the Internet of Things (IoT). These technologies help organizations improve production planning, inventory management, quality control, and supply chain efficiency. As a result, researchers have increasingly focused on studying the role of Business Intelligence in improving organizational performance and supporting data-driven decision-making.

Business Intelligence and Organizational Decision-Making:

Business Intelligence is widely recognized as a strategic tool that enhances the quality of business decisions. According to previous studies, BI provides managers with timely, accurate, and relevant information by integrating data from multiple organizational sources. Instead of relying on intuition or manual reports, managers can use analytical dashboards and reports to evaluate business performance and make evidence-based decisions.

Several researchers have highlighted that Business Intelligence supports decision-making at three organizational levels:

Strategic Level: Long-term planning, market analysis, and business growth.
Tactical Level: Resource allocation, departmental performance monitoring, and budgeting.
Operational Level: Daily production monitoring, inventory management, and quality improvement.
The literature suggests that organizations using BI systems can respond more quickly to market changes, improve forecasting accuracy, and reduce operational risks.

Business Intelligence in the Manufacturing Industry:

Manufacturing organizations generate enormous amounts of operational data through production equipment, ERP systems, quality inspections, inventory records, and supply chain activities. Earlier studies indicate that Business Intelligence enables manufacturers to transform this operational data into valuable business insights.

Researchers have found that BI tools assist manufacturing organizations in:

- Monitoring production performance in real time.
- Reducing machine downtime.
- Improving Overall Equipment Effectiveness (OEE).
- Identifying production bottlenecks.
- Optimizing inventory levels.
- Enhancing product quality.
- Supporting continuous improvement initiatives.

Many studies conclude that organizations implementing Business Intelligence experience increased productivity, reduced operational costs, and improved customer satisfaction.

Role of SQL, Python, and Power BI:

Modern Business Intelligence systems rely on various analytical tools for data management and visualization. Previous research highlights the importance of SQL, Python, and Microsoft Power BI in supporting business analytics.

SQL (Structured Query Language) is commonly used for extracting and managing structured data from organizational databases. Researchers emphasize that SQL improves data accessibility and enables efficient reporting through filtering, aggregation, and data retrieval.

Python has become one of the most popular programming languages for data analytics. Literature shows that Python supports data cleaning, preprocessing, exploratory data analysis (EDA), statistical analysis, and automation. Libraries such as Pandas, NumPy, and Matplotlib enable analysts to process large datasets efficiently.

Microsoft Power BI is recognized as one of the leading Business Intelligence platforms for creating interactive dashboards and reports. Researchers report that Power BI improves communication of analytical results through charts, KPI cards, slicers, and drill-down reports, enabling managers to understand business performance more effectively.

Together, these tools provide a comprehensive Business Intelligence environment that supports data-driven decision-making.

Business Intelligence and Operational Efficiency:

Operational efficiency is one of the most widely discussed benefits of Business Intelligence in existing literature. Researchers explain that BI helps organizations improve productivity by monitoring production processes, identifying inefficiencies, and optimizing resource utilization.

Several studies report that organizations using BI dashboards experience:

- Faster report generation.
- Reduced manual reporting effort.
- Improved production scheduling.
- Better utilization of machines and labor.
- Lower operational costs.
- Improved production planning.

Business Intelligence also enables organizations to monitor Key Performance Indicators (KPIs) continuously, allowing managers to take corrective actions before operational issues become significant.

Business Intelligence and Supply Chain Management:

The literature also emphasizes the growing role of Business Intelligence in supply chain management. Manufacturing organizations depend on efficient coordination between suppliers, production departments, warehouses, logistics providers, and customers.

Researchers suggest that BI supports supply chain performance by:

- Forecasting customer demand.
- Monitoring inventory availability.
- Evaluating supplier performance.
- Tracking order fulfillment.
- Optimizing logistics operations.
- Reducing inventory holding costs.

Real-time dashboards improve visibility across the supply chain and help managers respond quickly to disruptions, thereby improving customer service and operational efficiency.

Business Intelligence and Quality Management:

Quality management is another important area discussed in previous research. Studies indicate that Business Intelligence enables organizations to monitor quality-related KPIs, rejection rates, defect patterns, and customer complaints.

Through analytical dashboards and statistical reports, managers can identify the root causes of quality problems and implement corrective actions. Continuous monitoring improves product quality, reduces waste, and enhances customer satisfaction.

Researchers also note that integrating production data with quality information provides organizations with a comprehensive view of manufacturing performance, enabling continuous process improvement.

1. Gap Analysis:

The review of existing literature reveals that Business Intelligence has been widely studied across various industries, including manufacturing, healthcare, finance, retail, and logistics. Most previous studies focus on the general benefits of Business Intelligence, such as improved decision-making, enhanced operational efficiency, better reporting, and increased organizational performance. While these studies provide valuable insights, several research gaps still exist.

Firstly, many studies examine Business Intelligence from a broad organizational perspective without specifically focusing on its impact within the manufacturing industry. Since manufacturing organizations have unique operational processes, production systems, and performance requirements, there is a need for industry-specific research that evaluates how BI supports production planning, quality management, inventory control, and operational decision-making.

Secondly, previous research often emphasizes advanced Business Intelligence technologies but provides limited discussion on the practical use of commonly adopted analytical tools such as **SQL, Python, and Microsoft Power BI**. These tools are widely used by organizations for data extraction, cleaning, visualization, and reporting, making their practical application an important area for investigation.

Another gap identified in the literature is the limited availability of research focusing on **Indian manufacturing organizations**. Much of the existing research is based on developed countries, where technological infrastructure and BI adoption levels differ significantly from those in India. Therefore, studying Business Intelligence within the Indian manufacturing context can provide more relevant and practical insights.

Furthermore, many earlier studies concentrate primarily on the advantages of Business Intelligence while giving comparatively less attention to implementation challenges such as data quality, employee training, organizational resistance, system integration, and infrastructure limitations. Understanding these challenges is essential for improving the successful adoption of BI systems.

Based on these gaps, the present study aims to examine the **impact of Business Intelligence tools on organizational decision-making in the manufacturing industry**, with particular emphasis on the practical application of **SQL, Python, and Microsoft Power BI**. The study also evaluates both the benefits and challenges of Business Intelligence implementation and provides recommendations that can support manufacturing organizations in adopting effective data-driven decision-making practices.

CHAPTER 3

RESEARCH METHODOLOGY

1. Objectives of the Study:

The success of any research depends on clearly defined objectives that provide direction to the study. The present research aims to examine the impact of Business Intelligence (BI) tools on organizational decision-making in the manufacturing industry. As manufacturing organizations increasingly adopt digital technologies, Business Intelligence has become an essential tool for converting raw business data into meaningful information that supports effective decision-making. This study seeks to understand how BI contributes to operational efficiency, business performance, and strategic planning.

The specific objectives of the study are as follows:

- To understand the role of Business Intelligence tools in organizational decision-making within the manufacturing industry.
- To evaluate the impact of Business Intelligence on operational efficiency, productivity, and overall business performance.
- To examine the practical application of SQL, Python, and Microsoft Power BI in data extraction, analysis, visualization, and reporting.
- To identify the major benefits and challenges associated with the implementation of Business Intelligence systems.
- To provide recommendations that support data-driven decision-making and encourage the effective adoption of Business Intelligence technologies in manufacturing organizations.

These objectives form the foundation of the research and guide the process of data collection, analysis, interpretation, and conclusion.

1. Scope of the Study:

The scope of this research is limited to studying the impact of Business Intelligence tools on organizational decision-making within the manufacturing industry. The research primarily focuses on understanding how Business Intelligence enables organizations to improve operational efficiency, monitor business performance, and support strategic and operational decision-making.

Manufacturing organizations generate large amounts of data from production processes, inventory management, procurement activities, quality inspections, maintenance operations, finance, and supply chain management. This study investigates how Business Intelligence transforms this data into meaningful insights that assist managers in making informed business decisions.

The research emphasizes the practical application of commonly used Business Intelligence technologies, including SQL, Python, and Microsoft Power BI. SQL is used for extracting and managing structured data from organizational databases, Python supports data cleaning and analytical processing, while Microsoft Power BI is utilized for dashboard creation and business reporting. These technologies represent the core analytical tools used in modern Business Intelligence environments.

Geographically, the study mainly focuses on manufacturing organizations operating in India. However, references to global Business Intelligence practices are also considered to provide broader understanding and comparison with international developments. The study includes professionals working in production, quality, operations, business analytics, and management who are involved in organizational decision-making.

The study is confined to the managerial and business aspects of Business Intelligence implementation. It does not examine software development methodologies, database architecture, network infrastructure, cybersecurity mechanisms, or detailed programming techniques. Similarly, financial feasibility analysis and software vendor comparisons are beyond the scope of this research.

Overall, the study provides valuable insights into how Business Intelligence contributes to organizational effectiveness, operational excellence, and competitive advantage within manufacturing organizations.

2. Methodology:

Research methodology refers to the systematic process adopted to conduct scientific research and achieve the objectives of the study. It provides a structured framework for collecting, organizing, analyzing, and interpreting data in a logical and reliable manner. Selecting an appropriate research methodology is essential because it ensures that the findings are valid, objective, and relevant to the research problem.

The present study adopts a **mixed-method research approach**, combining both quantitative and qualitative methods. This approach provides a comprehensive understanding of the research topic by integrating numerical analysis with descriptive information. Quantitative methods help measure the effectiveness of Business Intelligence tools using survey responses and statistical analysis, while qualitative methods provide deeper insights into organizational practices, implementation challenges, and user experiences.

The research utilizes both **primary** and **secondary** sources of information. Primary data is collected directly from professionals working in manufacturing organizations through structured questionnaires. The respondents include production engineers, production managers, supervisors, business analysts, quality engineers, operations executives, and other employees involved in decision-making activities. Their responses provide practical insights regarding the implementation and effectiveness of Business Intelligence tools in manufacturing environments.

Secondary data plays an equally important role in supporting the research. Information is collected from academic journals, research articles, books, conference papers, industry reports, company publications, government reports, and reliable online databases. These sources help establish the theoretical foundation of the study and provide a broader understanding of Business Intelligence concepts, implementation strategies, and current industry trends.

The methodology also involves the practical application of analytical tools such as SQL, Python, Microsoft Excel, and Microsoft Power BI. SQL is used for extracting and organizing data from structured databases. Python assists in data cleaning, preprocessing, exploratory data analysis (EDA), and statistical analysis. Power BI is employed for creating interactive dashboards and visual reports that simplify complex business information and support managerial decision-making. Microsoft Excel is used for preliminary data organization and descriptive statistical analysis.

The collected data is carefully verified, organized, and classified according to the research objectives. Appropriate statistical techniques and visualization methods are then applied to identify trends, relationships, and patterns within the data. The findings are interpreted in relation to the research objectives to evaluate the impact of Business Intelligence on organizational decision-making.

The selected methodology ensures that the research findings are systematic, reliable, and relevant to current Business Intelligence practices within the manufacturing industry. By combining theoretical knowledge with practical analysis, the study provides meaningful insights into the role of Business Intelligence in improving organizational performance and supporting data-driven decision-making.

2.1. Research Design:

Research design refers to the overall framework used to collect, measure, and analyze data in order to answer the research questions effectively. It serves as a blueprint that guides the entire research process, ensuring that the study is systematic, reliable, and capable of achieving its objectives. Selecting an appropriate research design is essential because it determines the methods used for data collection, analysis, and interpretation.

The present study adopts a combination of **descriptive** and **exploratory** research designs.

Descriptive Research Design:

The descriptive research design is used to explain the current status of Business Intelligence adoption in manufacturing organizations. It focuses on describing how organizations implement Business Intelligence tools, how these tools support organizational decision-making, and the extent to which they improve operational efficiency.This research design enables the collection of factual information regarding various aspects of Business Intelligence, including:

- Production planning and monitoring
- Inventory management
- Supply chain performance
- Quality control

- Performance measurement
- Reporting accuracy
- Data visualization
- Decision-making effectiveness

Descriptive research provides a clear understanding of the existing practices followed by manufacturing organizations and evaluates their impact on business performance.

Exploratory Research Design:

The exploratory research design complements the descriptive approach by investigating areas where limited information is available. It helps identify emerging trends, implementation challenges, and future opportunities related to Business Intelligence in manufacturing.

The exploratory aspect of this research examines issues such as:

- Data quality challenges
- Integration of multiple information systems
- Employee adoption of BI tools
- Training requirements
- Cost of implementation
- Future role of Artificial Intelligence and predictive analytics

By combining descriptive and exploratory research designs, the study develops a comprehensive understanding of how Business Intelligence influences organizational decision-making. While the descriptive approach explains the current situation, the exploratory approach identifies opportunities for improvement and future developments.

2.2. Data Collection:

Data collection is one of the most important stages of the research process because the quality of research findings depends on the accuracy and reliability of the collected information. The present study uses both **primary** and **secondary** data sources to ensure comprehensive and balanced analysis.

Primary Data:

Primary data refers to information collected directly from respondents for the specific purpose of this research. A structured questionnaire was designed to gather information from professionals working in manufacturing organizations.

The respondents include:

- Production Engineers
- Production Managers
- Manufacturing Supervisors
- Quality Engineers
- Business Analysts
- Data Analysts
- Operations Executives
- Other professionals involved in organizational decision-making

The questionnaire contains both multiple-choice and Likert-scale questions that measure respondents' opinions regarding the use and effectiveness of Business Intelligence tools. The survey focuses on several important aspects, including:

- Frequency of Business Intelligence usage
- Types of BI tools used
- Effectiveness of SQL, Python, and Power BI
- Reporting efficiency
- Operational performance
- Decision-making speed
- Data accessibility
- Business performance improvements
- Challenges during BI implementation

The responses obtained from participants provide practical insights into how Business Intelligence supports organizational decision-making within manufacturing organizations.

Secondary Data:

Secondary data consists of information that has already been published by researchers, organizations, and institutions. It is collected to establish the theoretical foundation of the study and compare the findings with previous research.

The major sources of secondary data include:

- Research journals
- Books
- Conference proceedings
- Industry reports
- Company annual reports
- Government publications
- Business magazines
- Academic websites
- Reliable online databases

Secondary data helps understand the evolution of Business Intelligence, current industry trends, implementation strategies, best practices, and previous research findings. It also strengthens the credibility of the research by supporting primary data with existing academic literature.

The combination of primary and secondary data enables the study to present a balanced, comprehensive, and reliable analysis of Business Intelligence implementation in manufacturing organizations.

2.3. Sampling Method :

Sampling refers to the process of selecting a representative group of respondents from the target population. Since collecting information from every employee in the manufacturing industry is not practical, an appropriate sampling technique is necessary.

The present study uses **Purposive Sampling**, also known as **Judgmental Sampling**, which is a non-probability sampling technique.

Under this method, respondents are selected based on their knowledge, experience, and involvement in Business Intelligence implementation and organizational decision-making. Only individuals with relevant professional experience are included in the survey to ensure the reliability of the collected data.

The target respondents include:

- Production Managers
- Production Engineers
- Quality Engineers
- Manufacturing Supervisors
- Business Analysts
- Data Analysts
- Operations Executives
- Supply Chain Professionals

These professionals regularly use Business Intelligence tools or work with business data, making them suitable participants for this research.

Purposive sampling offers several advantages. It improves the relevance of the collected data, increases the quality of responses, and ensures that participants possess practical knowledge of Business Intelligence applications. Although the findings may not represent every manufacturing organization, they provide meaningful insights into current Business Intelligence practices.

2.4. Data Analysis Tools:

After data collection, the information is organized, processed, and analyzed using various Business Intelligence and analytical tools. These tools help convert raw data into meaningful information that supports the research objectives.

SQL (Structured Query Language):

SQL is used to retrieve, filter, organize, and manage structured data stored in relational databases. It enables efficient extraction of manufacturing data by using queries for sorting, grouping, joining tables, and generating summary reports. SQL ensures data accuracy and consistency before further analysis.

Python:

Python is used for data cleaning, preprocessing, exploratory data analysis (EDA), and statistical analysis. Python libraries such as Pandas, NumPy, and Matplotlib assist in handling missing values, identifying trends, performing descriptive statistics, and preparing datasets for visualization. Automation through Python also improves the efficiency of data analysis.

Microsoft Power BI:

Microsoft Power BI is used to create interactive dashboards, charts, KPI indicators, and visual reports. It helps transform complex datasets into easy-to-understand visualizations that support organizational decision-making. Managers can monitor production performance, identify operational bottlenecks, evaluate key metrics, and make informed business decisions using Power BI dashboards.

Microsoft Excel:

Microsoft Excel is used for preliminary data organization, data validation, descriptive statistics, and preparation of tables and charts. Excel serves as a supporting analytical tool before importing datasets into Power BI or Python for advanced analysis.

Together, these analytical tools provide a comprehensive framework for processing business data, identifying meaningful patterns, and generating insights that support data-driven decision-making in manufacturing organizations.

3. Period of Study:

The period of study refers to the duration during which the research activities, including data collection, analysis, interpretation, and report preparation, were carried out. Defining the study period is important because it establishes the timeframe within which the research findings are applicable and ensures that the data represents the current business environment.

The present study was conducted over a period of **eight weeks, from May 2026 to June 2026**. During this period, both primary and secondary data were collected and analyzed to achieve the objectives of the research.

The research process was carried out in several stages. Initially, an extensive review of books, research journals, conference papers, company reports, and industry publications was undertaken to understand the theoretical concepts of Business Intelligence and its applications in the manufacturing industry. This helped establish the conceptual framework for the study and identify existing research gaps.

Following the literature review, a structured questionnaire was prepared to collect primary data from professionals working in manufacturing organizations. The questionnaire was distributed to respondents who possessed practical knowledge of Business Intelligence tools and organizational decision-making processes. After collecting the responses, the data was carefully verified, organized, and prepared for analysis.

The collected data was then analyzed using SQL, Python, Microsoft Excel, and Microsoft Power BI. Various analytical techniques, descriptive statistics, and visualization methods were applied to identify trends, relationships, and performance indicators. Finally, the findings were interpreted and compared with existing literature to draw meaningful conclusions and provide recommendations.Although the research was conducted within a limited time period, every effort was made to ensure that the collected information was accurate, reliable, and relevant to the objectives of the study.

4. Limitations of the Study:

Every research study has certain limitations that may influence the interpretation and generalization of its findings. Identifying these limitations enhances the transparency and credibility of the research while providing direction for future studies.

One of the primary limitations of this study is that it is confined to the **manufacturing industry**. Therefore, the findings may not be directly applicable to organizations operating in other sectors such as healthcare, banking, retail, education, or information technology.

Another limitation is the relatively **limited sample size**. Due to time and accessibility constraints, data was collected from a selected group of professionals rather than from all manufacturing organizations. Although purposive sampling ensured that respondents possessed relevant knowledge and experience, the findings may not fully represent the opinions of the entire manufacturing sector.

The research also relies on **self-reported responses** collected through questionnaires. Since respondents expressed their opinions and experiences based on personal observations, there is a possibility of response bias or subjective interpretation. However, efforts were made to design clear and unbiased questions to minimize this limitation.

The study focuses specifically on the practical application of **SQL, Python, Microsoft Power BI, and Business Intelligence concepts**. Other analytical platforms, cloud-based BI solutions, and advanced artificial intelligence applications have not been examined in detail. Future studies may explore these technologies to provide a broader understanding of Business Intelligence implementation.

Another limitation relates to the **time available for conducting the research**. The study was completed within an eight-week period, which restricted the duration of primary data collection and detailed organizational observation. A longer research period could provide more extensive data and deeper analysis.

Finally, the study emphasizes the managerial and operational aspects of Business Intelligence rather than its technical implementation. Areas such as software architecture, database optimization, cybersecurity, cloud infrastructure, and programming methodologies are beyond the scope of this research.

Despite these limitations, the study provides meaningful insights into the role of Business Intelligence in improving organizational decision-making and operational performance within manufacturing organizations.

5. Utility of Research:

The findings of this research are expected to provide valuable academic as well as practical contributions to various stakeholders involved in Business Intelligence and manufacturing management.

From an organizational perspective, the study helps manufacturing companies understand how Business Intelligence tools can improve decision-making by transforming raw business data into meaningful information. Managers can use the research findings to strengthen production planning, monitor operational performance, optimize inventory, improve quality management, and enhance overall organizational efficiency.

The study also demonstrates the practical application of modern analytical tools such as **SQL, Python, Microsoft Power BI, and Microsoft Excel**. Organizations seeking to adopt data-driven decision-making can use these findings as a reference for implementing Business Intelligence solutions and developing effective reporting systems.

For business managers and decision-makers, the research highlights the importance of real-time dashboards, key performance indicators (KPIs), and data visualization in improving business performance. The recommendations provided in this study may support better planning, resource allocation, performance monitoring, and strategic decision-making.

The research is equally beneficial for students, academicians, and future researchers in the fields of Business Intelligence, Data Analytics, Operations Management, and Manufacturing Management. It provides a comprehensive understanding of Business Intelligence concepts, implementation strategies, benefits, and challenges, thereby contributing to the existing body of academic literature.

Furthermore, the study identifies several implementation challenges such as data quality, employee training, system integration, and organizational resistance. Understanding these challenges enables organizations to develop more effective implementation strategies and maximize the benefits of Business Intelligence investments.

The findings may also assist policymakers and industry professionals in promoting digital transformation within the manufacturing sector. As organizations increasingly adopt Industry 4.0 technologies, Business Intelligence will continue to play a vital role in improving competitiveness, innovation, and sustainable business growth.

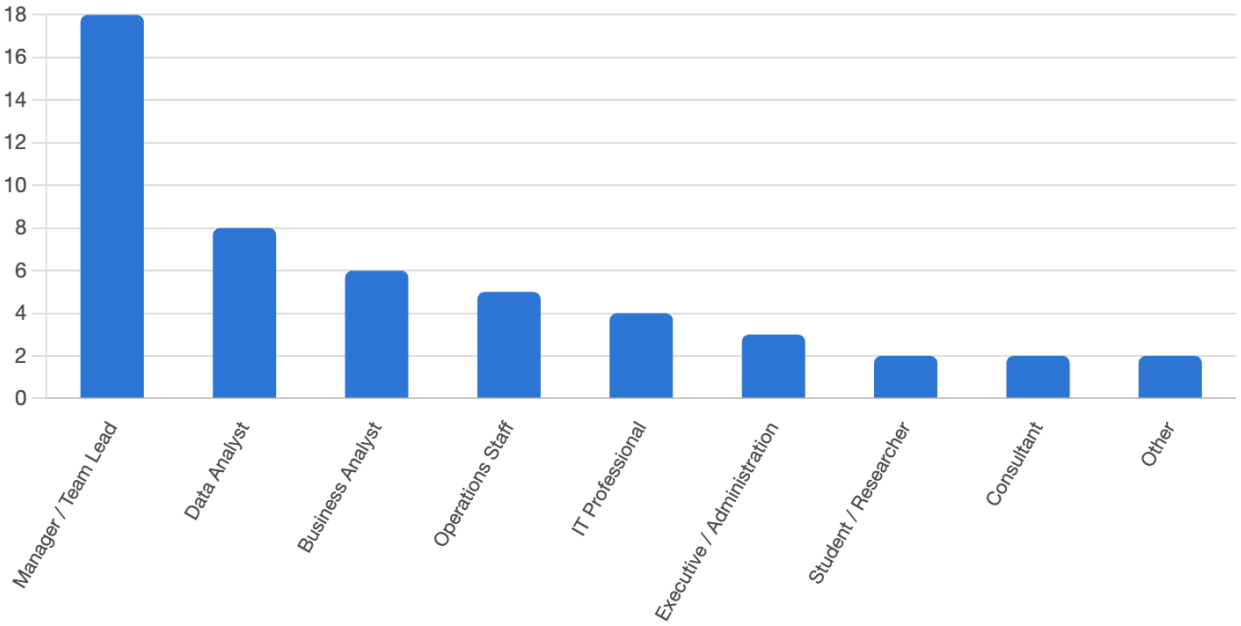
In conclusion, this research serves as a useful reference for manufacturing organizations, researchers, students, and professionals seeking to understand the impact of Business Intelligence on organizational decision-making. By demonstrating the practical use of SQL, Python, and Power BI, the study contributes to the growing importance of data-driven management and supports the successful implementation of Business Intelligence practices in the manufacturing industry.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

Table 4.1: Distribution of Respondents by Job Role:

Job Role	Frequency	Percentage (%)
Manager / Team Lead	18	36%
Data Analyst	8	16%
Business Analyst	6	12%
Operations Staff	5	10%
IT Professional	4	8%
Executive / Administration	3	6%
Student / Researcher	2	4%
Consultant	2	4%
Other	2	4%
Total	**50**	**100%**



Analysis:

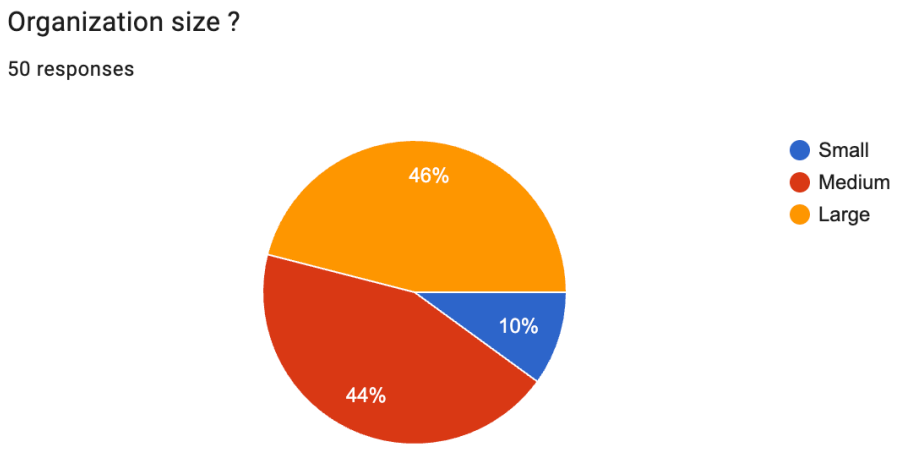
The survey included **50 respondents** from various professional backgrounds. The largest group of respondents were **Managers/Team Leads (36%)**, followed by **Data Analysts (16%)** and **Business Analysts (12%)**. Operations Staff accounted for **10%**, while IT Professionals represented **8%**. Executive/Administration staff contributed **6%**, and Students/Researchers, Consultants, and Others each represented **4%** of the respondents.

Interpretation

The survey responses are primarily from professionals involved in managerial and analytical roles. This indicates that the findings are based largely on individuals who are likely to use Business Intelligence tools for strategic planning, reporting, and decision-making. The diverse representation of job roles also provides a balanced perspective on the adoption and effectiveness of BI tools across different organizational functions.

Table 4.2: Distribution of Respondents by Organization Size:

Organization Size	Frequency	Percentage (%)
Large	23	46%
Medium	22	44%
Small	5	10%
Total	**50**	**100%**



Analysis:

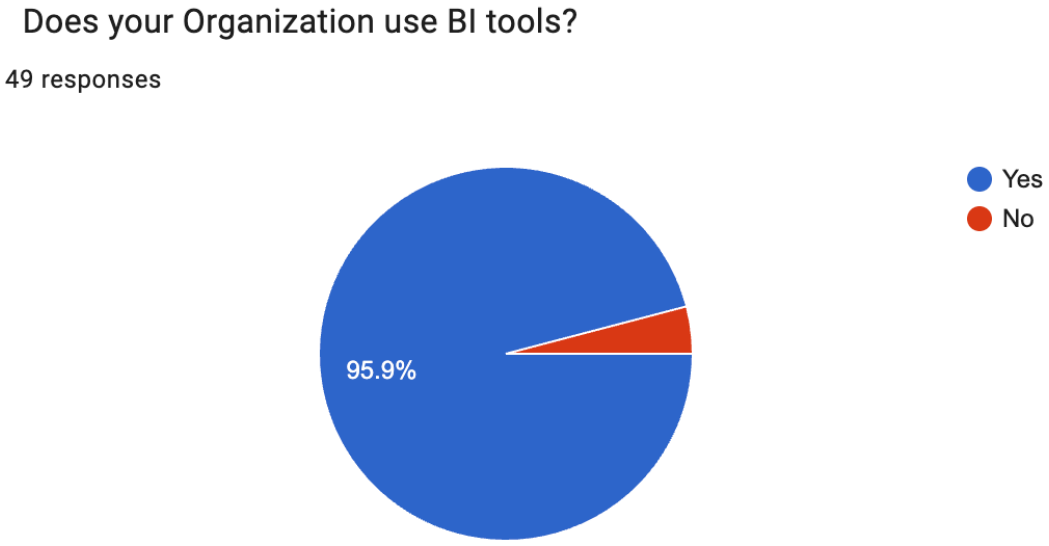
Out of the 50 respondents, 23 (46%) belong to large organizations, 22 (44%) work in medium-sized organizations, and 5 (10%) are from small organizations. The responses are almost equally distributed between medium and large organizations, while only a small proportion of respondents represent small organizations.

Interpretation:

The survey findings mainly reflect the perspectives of employees working in medium and large organizations, where Business Intelligence (BI) systems are more commonly implemented due to greater data management needs and organizational complexity. Since only 10% of respondents are from small organizations, the results are more representative of medium- and large-scale enterprises.

Table 4.3: Does Your Organization Use BI Tools?

Response	Frequency	Percentage (%)
Yes	47	94%
No	2	4%
No Response	1	2%
Total	**50**	**100%**



Analysis:

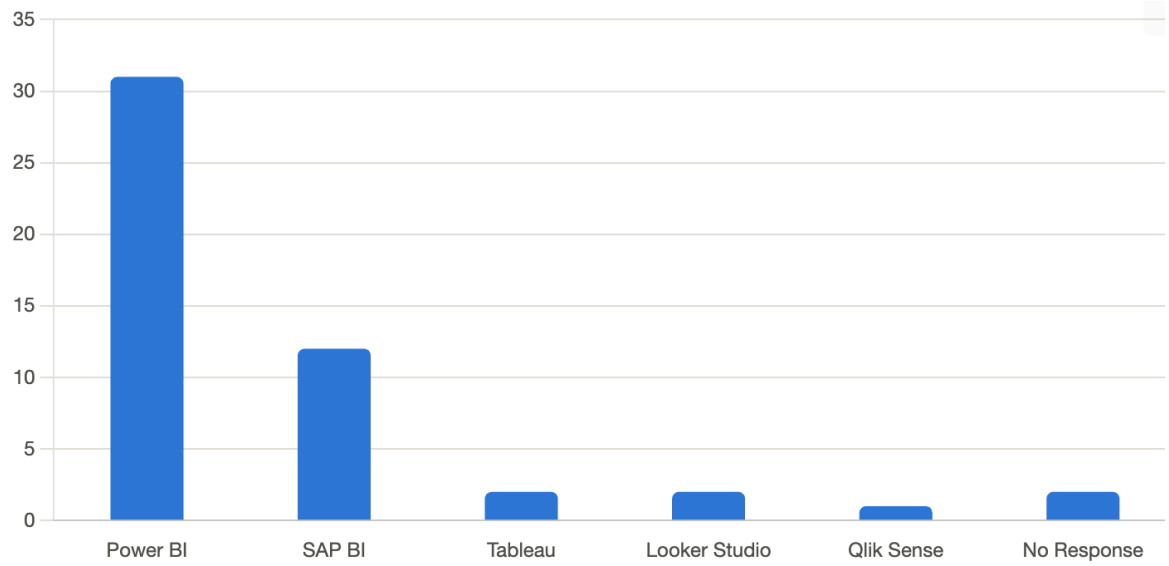
Out of the 50 respondents, 47 (94%) stated that their organizations use Business Intelligence (BI) tools, while 2 (4%) reported that their organizations do not use BI tools. One respondent (2%) did not provide a response.

Interpretation:

The findings indicate that Business Intelligence tools are widely adopted, with an overwhelming majority of respondents confirming their use within their organizations. This suggests that BI has become an essential component of organizational decision-making and performance management. The very small percentage of organizations not using BI tools highlights the growing importance of data-driven practices across industries.

Table 4.4: Business Intelligence (BI) Tools Used in the Organization:

BI Tool	Frequency	Percentage (%)
Power BI	31	62%
SAP BI	12	24%
Tableau	2	4%
Looker Studio	2	4%
Qlik Sense	1	2%
No Response	2	4%
Total	**50**	**100%**



Analysis:

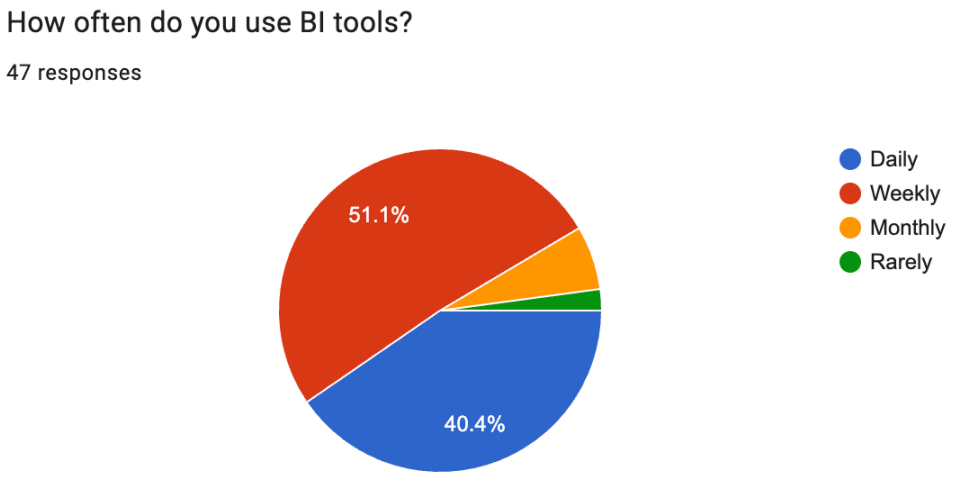
Among the 50 respondents, Power BI is the most widely used Business Intelligence tool, with 31 respondents (62%) reporting its use. SAP BI is the second most commonly used tool, accounting for 12 respondents (24%). Tableau and Looker Studio are each used by 2 respondents (4%), while Qlik Sense is used by 1 respondent (2%). Additionally, 2 respondents (4%) did not specify the BI tool used in their organization.

Interpretation:

The results indicate that Power BI is the dominant Business Intelligence platform among the surveyed organizations, likely due to its user-friendly interface, integration with Microsoft products, and cost-effectiveness. Although SAP BI also has a significant presence, other BI tools such as Tableau, Looker Studio, and Qlik Sense have comparatively lower adoption. This suggests that Power BI is the preferred choice for data visualization and business analytics in most organizations represented in the survey.

Table 4.5: Frequency of Using Business Intelligence (BI) Tools:

Usage Frequency	Frequency (f)	Percentage (%)
Weekly	24	51.06%
Daily	19	40.43%
Monthly	3	6.38%
Rarely	1	2.13%
Total Valid Responses	**47**	**100.00%**



Analysis:

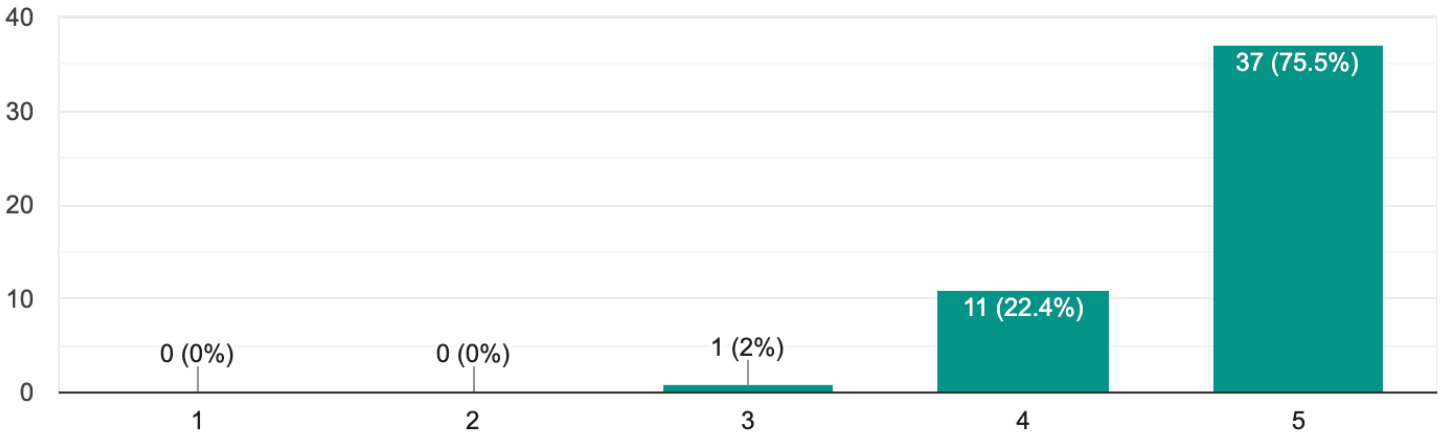
The survey results show that 24 respondents (51%) use Business Intelligence tools on a daily basis, making it the most common frequency of use. 19 respondents (40.4%) use BI tools weekly, while 3respondents (6.38%) access them monthly. Only 1 respondents (2.1%) reported using BI tools rarely.

Interpretation:

The findings indicate that Business Intelligence tools are an integral part of the daily operations of many organizations. Nearly half of the respondents rely on BI tools every day, demonstrating their importance in monitoring performance, generating reports, and supporting data-driven decision-making. The relatively small percentage of respondents using BI tools rarely suggests that these tools have become a regular and valuable resource in most workplaces.

Table 4.6: Business Intelligence (BI) Improves Organizational Decision-Making:

Likert Scale Rating	Frequency (f)	Percentage (%)
5 - Strongly Agree	37	75.51%
4 - Agree	11	22.45%
3 - Neutral	1	2.04%
2 - Disagree	0	0.00%
1 - Strongly Disagree	0	0.00%
Total Valid Responses	49	100.00%



Analysis:

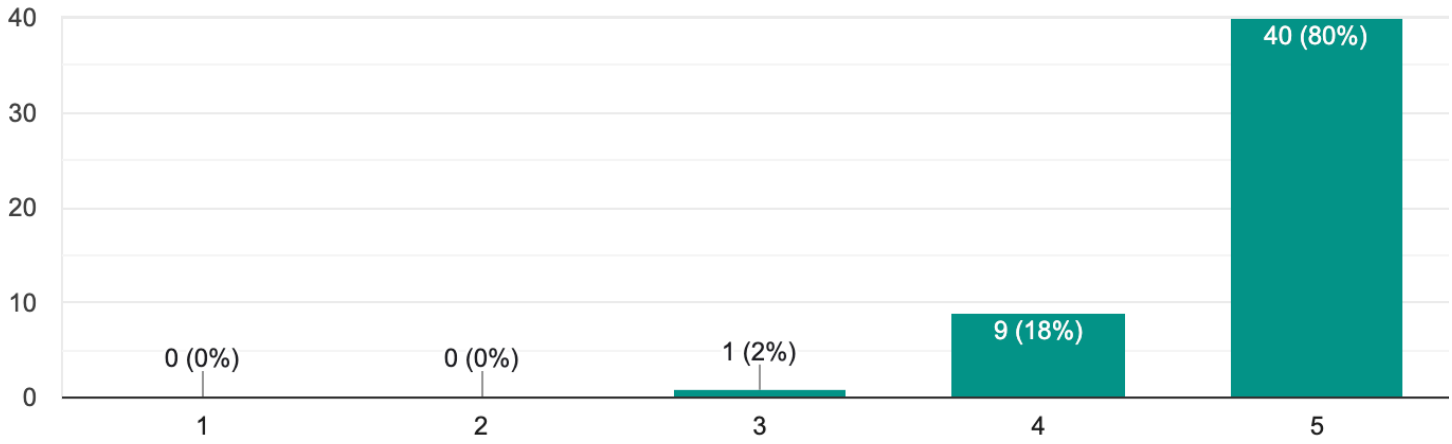
The table shows that **97.96%** of respondents either **agreed or strongly agreed** that Business Intelligence tools improve organizational decision-making.While **2.04%** remained neutral, 0**%** expressed disagreement.

Interpretation:

The findings indicate that Business Intelligence has a positive impact on organizational decision-making by providing accurate and timely information, enabling managers to make more informed and effective decisions.

Table 4.7: Business Intelligence (BI) Improves Operational Efficiency:

Likert Scale Rating	Frequency (f)	Percentage (%)
5 - Strongly Agree	40	80.00%
4 - Agree	9	18.00%
3 - Neutral	1	2.00%
2 - Disagree	0	0.00%
1 - Strongly Disagree	0	0.00%
Total	50	100.00%



Analysis

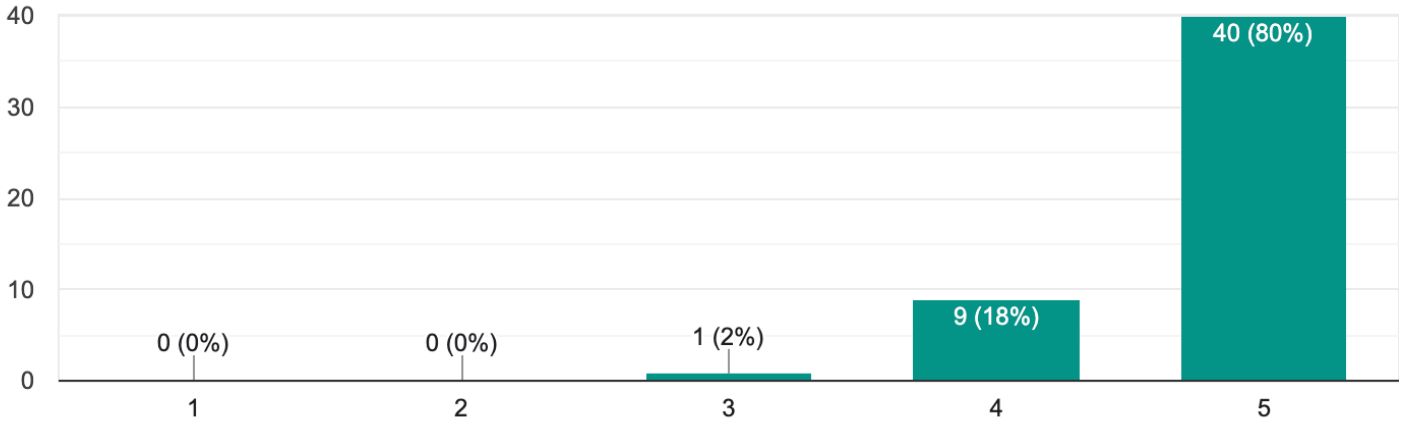
The table shows that **98%** of respondents agreed or strongly agreed that Business Intelligence improves operational efficiency. **2%** of respondents were neutral, while **0%** disagreed.

Interpretation

The findings indicate that Business Intelligence tools help improve operational efficiency by streamlining processes, reducing manual work, and enabling better monitoring of business performance.

Table 4.8: Business Intelligence (BI) Increases Employee Productivity:

Response	Frequency	Percentage (%)
Strongly Agree	40	80.00%
Agree	9	18.00%
Neutral	1	2.00%
Disagree	0	0.00%
Strongly Disagree	0	0.00%
Total	50	100.00%



Analysis:

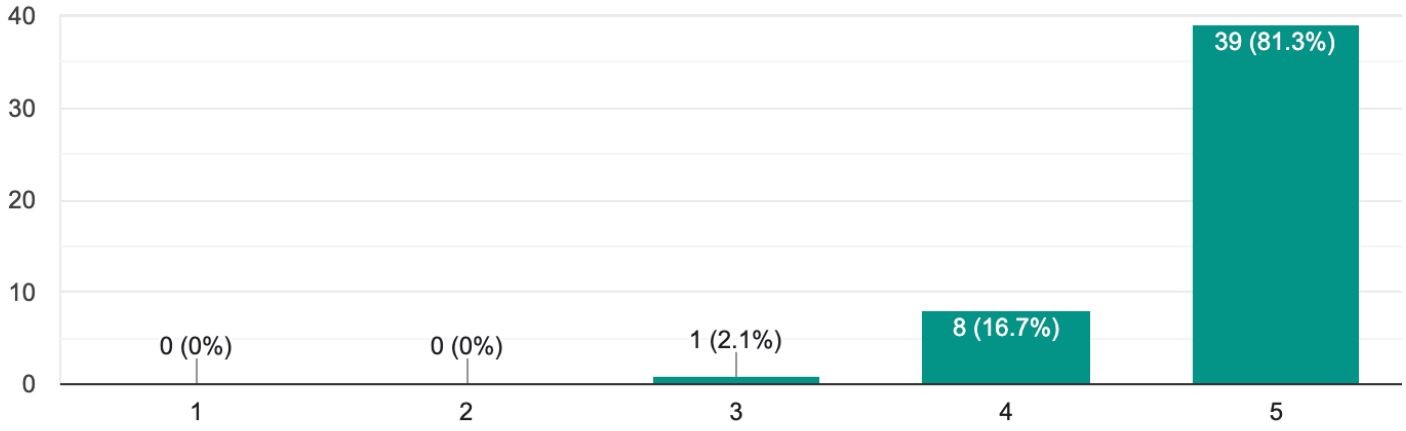
The table shows that **98%** of respondents agreed or strongly agreed that Business Intelligence tools increase employee productivity. 2**%** remained neutral, while **0%** disagreed.

Interpretation:

The findings suggest that Business Intelligence tools improve employee productivity by simplifying data analysis, reducing manual reporting, and enabling faster access to information, leading to more efficient work processes.

Table 4.9: Business Intelligence (BI) Reports are Easy to Understand:

Response	Frequency	Percentage (%)
Strongly Agree	39	81.25%
Agree	8	16.67%
Neutral	1	2.08%
Disagree	0	0
Strongly Disagree	0	0
Total	**50**	**100**



Analysis:

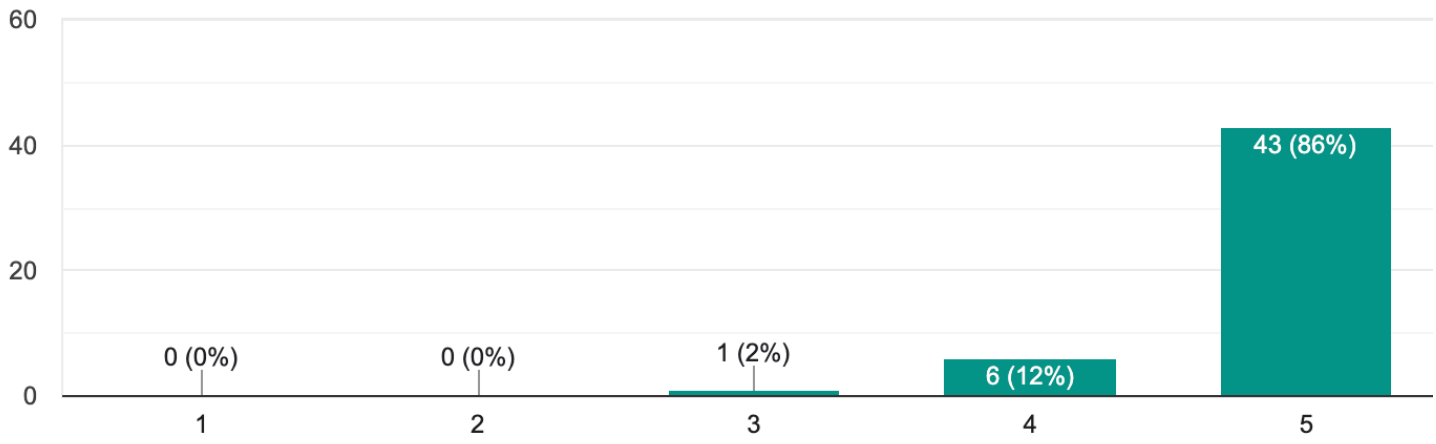
The table shows that **97.92%** of respondents agreed or strongly agreed that Business Intelligence reports are easy to understand. **2.08%** remained neutral, while only **0%** disagreed.

Interpretation:

The findings indicate that Business Intelligence tools provide clear and user-friendly reports and dashboards, enabling employees and managers to interpret data easily and make informed business decisions.

Table 4.10: Business Intelligence (BI) Helps Identify Business Trends and Patterns:

Response	Frequency	Percentage (%)
Strongly Agree	43	86.00%
Agree	6	12.00%
Neutral	1	2.00%
Disagree	0	0
Strongly Disagree	0	0
Total	**50**	**100.00%**



Analysis:

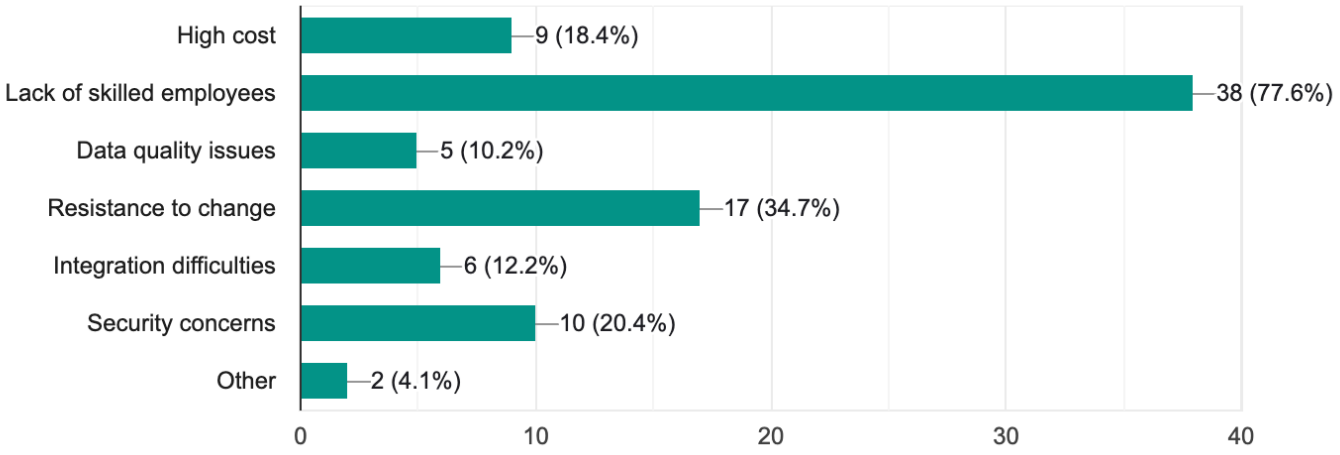
The table indicates that **98%** of respondents agreed or strongly agreed that Business Intelligence tools help identify business trends and patterns. **2%** of respondents remained neutral, while **0%** disagreed.

Interpretation:

The findings suggest that Business Intelligence tools are effective in identifying business trends and patterns, enabling organizations to make informed decisions, anticipate market changes, and improve strategic planning.

Table 4.11: Challenges Faced During Business Intelligence (BI) Implementation:

Major Challenge	Frequency	Percentage (%)
Data Quality Issues	5	10.2%
Lack of Skilled Employees	38	77.6%
High Implementation Cost	9	18.40%
Data Integration Issues	6	12.2%
Resistance to Change	17	34.7%
Total	**50**	**100**



Analysis:

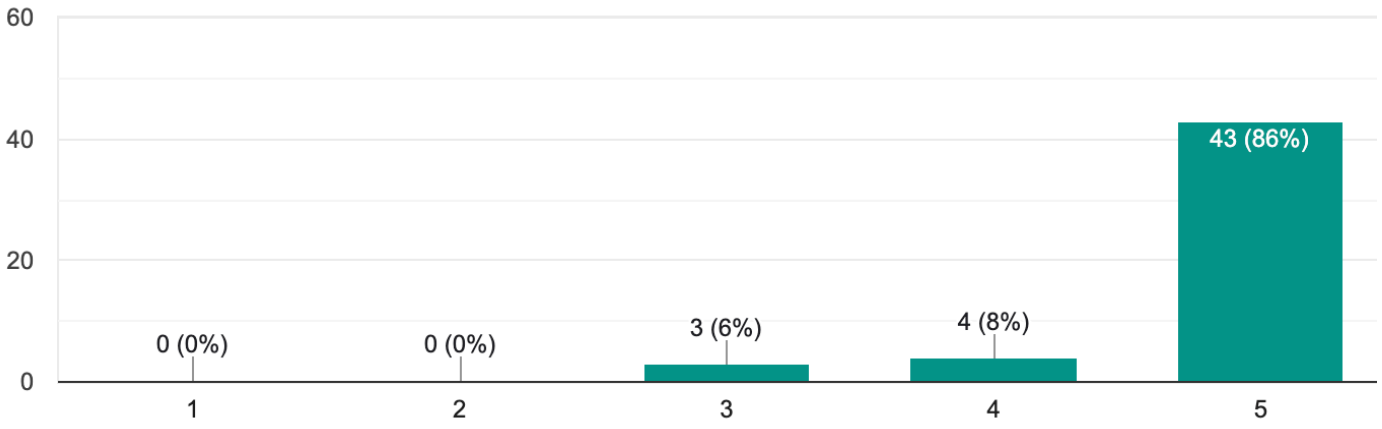
The table shows that **Lack of Skilled Employees (77.6%)** are the most common challenge faced during Business Intelligence implementation, followed by **Resistance to Change(34.7%)** and **High Implementation Cost (18.4%)**. Other challenges include Data Integration Issues (12.2%) and Data Quality Issues (10.2%).

Interpretation:

The findings indicate that although Business Intelligence offers significant benefits, organizations face challenges on workforce upscaling, training programs, and change management strategies to complement technical implementation..Addressing these issues can improve the successful adoption and effectiveness of BI systems.

Table 4.12: Overall Satisfaction with Business Intelligence (BI) Tools:

Response	Frequency	Percentage (%)
Very Satisfied	43	86.0%
Satisfied	4	8.0%
Neutral	3	6.0%
Dissatisfied	0	0.0%
Very Dissatisfied	0	0.0%
Total	**50**	**100.0%**



Analysis:

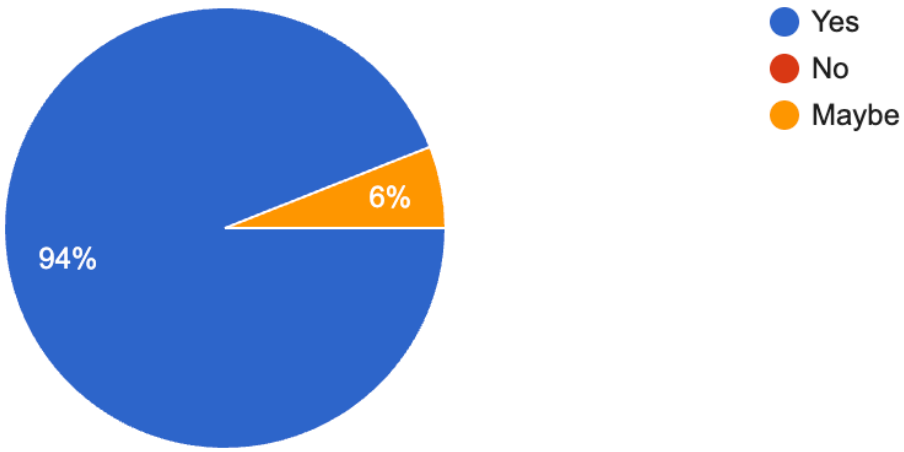
The table shows that **92%** of respondents are either **satisfied or very satisfied** with the use of Business Intelligence tools in their organizations. **6%** expressed a neutral opinion, while **0%** were dissatisfied or very dissatisfied.

Interpretation:

The findings indicate a high level of satisfaction among respondents regarding the use of Business Intelligence tools. This suggests that BI systems effectively support data analysis, reporting, and decision-making, contributing to improved organizational performance and user confidence.

Table 4.13: Recommendation to Adopt Business Intelligence (BI) Tools:

Response	Frequency	Percentage (%)
Yes	47	94.0%
No	3	6.0%
Total	**50**	**100.0%**



Analysis:

The table shows that **94%** of respondents would recommend the adoption of Business Intelligence tools in organizations, while **6%** would not recommend them.

Interpretation:

The findings indicate a strong positive perception of Business Intelligence tools among the respondents. The high recommendation rate suggests that BI systems are considered valuable for improving decision-making, operational efficiency, and overall organizational performance.

CHAPTER 5

FINDINGS, RECOMMENDATIONS AND CONCLUSION

Research Method:

This study is based on **primary research**. The data was collected using a **structured questionnaire through Google Forms** from **50 respondents** working in manufacturing organizations. The responses were analyzed using frequency and percentage analysis to evaluate the impact of Business Intelligence (BI) tools on organizational decision-making.

5.1 Findings Based on Observations:

1. Most respondents were employed in medium and large manufacturing organizations.
2. Production TeamLead and Production Managers formed the largest group of respondents.
3. Microsoft Power BI was the most widely used Business Intelligence tool.
4. Most respondents reported using BI tools on a daily or weekly basis.
5. Organizations using BI tools showed better reporting and monitoring practices.
6. Interactive dashboards were commonly used for tracking business performance.
7. Respondents demonstrated a positive attitude toward adopting Business Intelligence technologies.
8. Data quality and employee skill gaps were commonly observed challenges during BI implementation.

5.2 Findings Based on Analysis of Data:

1. A majority of respondents confirmed that their organizations use Business Intelligence tools.
2. Most respondents agreed that BI improves organizational decision-making by providing timely and accurate information.
3. The survey indicated that BI enhances operational efficiency by reducing manual reporting and improving process monitoring.

4. Respondents believed that Business Intelligence increases employee productivity and supports better resource utilization.
5. Most participants found BI reports and dashboards easy to understand and interpret.
6. Business Intelligence was considered effective in identifying business trends and supporting strategic planning.
7. Data quality issues, implementation costs, and lack of skilled professionals were identified as the major challenges.
8. Overall satisfaction with Business Intelligence tools was high, and most respondents recommended adopting BI solutions.

5.3 General Findings:

1. Business Intelligence has become an important component of modern manufacturing organizations.
2. Data-driven decision-making is increasingly replacing traditional decision-making approaches.
3. Microsoft Power BI is the preferred BI platform due to its ease of use and visualization capabilities.
4. Organizations recognize the value of BI in improving productivity and business performance.
5. Successful BI implementation depends on quality data, employee skills, and management support.
6. Organizations investing in Business Intelligence gain better operational control and competitive advantage.

5.4 Recommendations Based on Findings:

1. Manufacturing organizations should expand the adoption of Business Intelligence tools across all departments.
2. Organizations should provide regular training on SQL, Python, Power BI, and data analytics.
3. Companies should improve data quality through proper validation and governance practices.
4. BI systems should be integrated with ERP and other enterprise applications for seamless reporting.
5. Organizations should monitor Key Performance Indicators (KPIs) using interactive dashboards.
6. Management should encourage a culture of data-driven decision-making.

7. Organizations should periodically evaluate BI performance and update analytical tools as business needs evolve.
8. Small and medium-sized enterprises should adopt cost-effective BI solutions to improve operational efficiency.

5.5 Suggestions for Areas of Improvement:

1. Improve data accuracy and consistency before analysis.
2. Strengthen employee training on Business Intelligence tools.
3. Enhance integration between different organizational information systems.
4. Increase awareness of the benefits of Business Intelligence across all departments.
5. Upgrade existing BI infrastructure to support real-time analytics.
6. Improve cybersecurity measures to protect business data.
7. Encourage continuous monitoring and evaluation of BI performance.
8. Invest in advanced analytics and predictive reporting capabilities.

5.6 Scope for Future Research:

The present study focuses on the impact of Business Intelligence tools on organizational decision-making in the manufacturing industry using primary data collected through a Google Forms survey. Future research can be conducted with a larger sample size covering multiple industries such as healthcare, banking, retail, logistics, and information technology to compare the effectiveness of Business Intelligence across different sectors. Researchers may also examine the integration of Artificial Intelligence, Machine Learning, Predictive Analytics, and Cloud-Based Business Intelligence with modern BI platforms. Longitudinal studies can further evaluate the long-term impact of Business Intelligence on organizational performance, innovation, and competitive advantage.

5.7 Conclusion:

Business Intelligence has emerged as a critical technology for supporting organizational decision-making in the manufacturing industry. Based on the primary survey conducted through Google Forms, the findings indicate that Business Intelligence tools significantly improve decision-making, operational efficiency, employee productivity, and overall business performance. Microsoft Power BI was identified as the most widely used BI platform due to its interactive dashboards, reporting capabilities, and ease of use.

Although organizations face challenges such as data quality issues, implementation costs, and the shortage of skilled professionals, the overall perception of Business Intelligence remains highly positive. The study concludes that organizations adopting Business Intelligence technologies are better equipped to make informed decisions, optimize business operations, and achieve sustainable growth. Continued investment in Business Intelligence tools, employee training, and advanced analytics will further strengthen organizational performance and competitiveness in the future.

ANNEXURE

Customer Satisfaction Survey Questionnaire

- What is your job role?
- Organization size ?
- Does your Organization use BI tools?
- Which BI tools are used in your Organization?
- How often do you use BI tools?
- Which BI tools improve often use for decision-making?
- Is BI systems improve operational efficiency?
- Is BI tools increase productivity?
- Is BI reports are easy to understand?
- Is BI tools help identify business trends effectively?
- What challenges are faced during BI implementation?

Google form link: https://forms.gle/icGh2cGcyDHGjzeRA

Duplichecker

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